ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

SECTION 01 | PROFILE & DIRECTION

Self Introduction: Written Script and One-Page Profile

One-Page Profile

Harshitha S | First-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering (2026-2030). Interested in AI applications and agent engineering.

Current skills: Python and SQL programming; basic knowledge of C; communication, leadership, collaboration, and an active approach to learning.

Career direction: Build useful AI applications and agents that address practical needs. Strengthen computing fundamentals, software development, and responsible AI skills through coursework and projects.

Professional Introduction Script

Hello, I'm Harshitha S, a first-year Computer Science and Engineering student at St. Joseph's College of Engineering. I am interested in AI application and agent engineering, and I am building my skills in software development and generative AI.

I chose Computer Science because I am curious about technology and how data and software can address real-world problems. I have programming skills in Python and SQL, along with basic knowledge of C. I also bring communication and leadership skills that help me work with people, share ideas, and take responsibility in a team.

My goal is to strengthen my technical foundation, work on practical projects, and gain experience building AI applications and agents. I want to keep learning, test my ideas carefully, and contribute effectively to collaborative projects. Thank you for the opportunity to introduce myself.

English Learning Snapshot

Preply English Test: C1 - Advanced; 4,500 words known at this level and 1,250 remaining to the next level. **Speak & Improve:** Read Aloud #1 B1 baseline, checked 3 October 2026. **Reading practice screenshot:** 5 tests, 6 minutes total, 324 WPM average, 63% average comprehension.

SECTION 02 | IMPRESSIVE PERSONALITY

Jerry Liu

Co-founder and CEO of LlamaIndex | AI application builder | Technology entrepreneur

Introduction

Jerry Liu is a computer scientist and entrepreneur, best known as the co-founder and chief executive officer of LlamaIndex. The company builds tools that help developers connect large language models (LLMs) with data and create AI applications and agents. His work addresses a practical challenge in applied AI: models need relevant, specialized, and changing information to be useful.

Academic and Technical Background

Liu's public LinkedIn profile lists Princeton University. In a first-person account of LlamaIndex's early development, he describes following generative models through his academic and professional career, including research on generative adversarial networks (GANs) and sensor compression. This background links machine-learning ideas with the software systems needed to use them in applications.

The Beginning of LlamaIndex

The idea grew from a practical problem: Liu wanted a language model to help prepare for customer meetings using information in internal tools and call notes. The model did not have that company-specific context. Early experiments connected data sources to language models through a project called GPT Index, which later became LlamaIndex. Liu worked with Simon Suo to develop it into a broader open-source framework.

Professional Career and Contributions

As co-founder and CEO, Liu has helped guide LlamaIndex from an open-source framework for working with data and LLMs toward a broader set of developer tools and services. These include data ingestion and parsing, retrieval, question answering over private data, and agent workflows that use documents and tools.

A central pattern is to load information from files or other sources, organize and index it, retrieve relevant context, and provide that context to a model when answering. This retrieval-augmented generation (RAG) approach connects model capabilities with useful information. LlamaIndex's product direction has broadened toward document processing and agent workflows; the company announced a spreadsheet agent in 2025. In 2023 it announced an $8.5 million seed round led by Greylock.

SECTION 02 | IMPRESSIVE PERSONALITY

Impact, Qualities and Inspiration

A student reflection based on Jerry Liu's public work

Personality and Professional Qualities

A public career cannot establish private character. These qualities are interpretations of Liu's published work and product decisions:

• **Curiosity:** exploring how language models can work with new and specialized information.

• **Practical problem-solving:** turning a work need into a reusable developer tool.

• **Technical breadth:** combining machine learning, data systems, retrieval, and application design.

• **Adaptability:** extending data indexing toward agent workflows as application needs evolved.

• **Builder mindset:** sharing open-source tools for developers to inspect and experiment with.

Recognition and Impact

In June 2023, Liu reported that LlamaIndex had reached 16,000 GitHub stars and 200,000 monthly downloads within six months. These historical figures illustrate early developer interest in tools for data-aware AI.

Why Jerry Liu Is an Impressive Personality to Me

• He identified a real need: models require useful, reliable context.

• He connects technical engineering with product thinking.

• His work highlights data, retrieval, evaluation, and workflows alongside models.

• The evolution of LlamaIndex reflects continuous learning and adaptation.

• His example motivates me to build around real user needs.

Key Takeaway

Jerry Liu's journey reflects curiosity, practical problem-solving, technical depth, and adaptability. His work reminds aspiring AI engineers to look beyond the model and build the complete system that makes AI useful.

Sources and Further Reading

[1] Jerry Liu, public LinkedIn profile. [2] “Building the data framework for LLMs,” LlamaIndex (2023). [3] Latent Space interview, “RAG Is A Hack.” [4] LlamaIndex, “Introducing LlamaCloud and LlamaParse” (2024). [5] LlamaIndex, “Introducing the Spreadsheet Agent” (2025). [6] LlamaIndex and Weaviate materials on RAG (2023). Professional qualities are interpretations of public work, not claims about private character.

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Week 1: Course Introduction and Growth Mindset

The course opens with two preview lectures that recap an earlier course, “Developing Soft Skills”. That course dealt with perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The big new topic here is mindset: a person’s habitual attitudes and ways of reading a situation.

In the shoe-salesman story, nobody wears shoes. Is that “no chance” or a “splendid opportunity”? The same situation draws opposite responses. Drawing on Carol Dweck, the lectures set a fixed mindset against a growth mindset. People with a fixed mindset avoid challenges, resist change, dwell on their limits, reject criticism and stop learning once school is over. Growth-mindset people welcome challenges, learn from failure and treat learning as lifelong.

Learning mindsets form early. A belief like “I’m not smart enough” limits a person, but it isn’t a fact about their ability. The suggested “secrets” are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

Week 2: Time Management, Procrastination and Assertiveness

Time can’t really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for “time leakage” (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps. Covey’s urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and “time stealers” such as waiting and travel can be turned into “time gifters.”

The lectures trace procrastination to Parkinson’s law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the “frog” (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying “yes” when you mean “no.”

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else's, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy's commandments and learning four ways to say no.

On anger, Aristotle's point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln's unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a "window" that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely), Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can't buy.

Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character (“when character is lost, everything is lost”). Professional ethics combine written codes with society’s moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi’s line that the earth provides enough for everyone’s need but not everyone’s greed.

SECTION 04 | CAREER SWOC ROADMAP

SWOC Career Roadmap

Career Vision

Build the technical and professional capability to create useful, reliable AI applications and agent-based systems that solve practical problems.

Starting Point

I am a first-year Computer Science and Engineering student at St. Joseph's College of Engineering. I have programming skills in Python and SQL, basic knowledge of C, good communication and leadership skills, and a strong interest in learning.

Strengths

• Python and SQL foundation; introductory C knowledge.

• Communication, leadership, collaboration, curiosity, and active learning.

Weaknesses to Develop

• AI, machine-learning, and LLM concepts are still developing.

• I need a portfolio of applied AI projects and more practice with APIs, Git, testing, and deployment.

• I have limited field experience as a first-year student.

Opportunities

• Four-year academic runway, project-led learning, coding clubs, open source, workshops, and internships.

• Campus and everyday needs can become useful project prompts.

Challenges

• AI tools change quickly; internships are competitive; reliability, privacy, security, and time management need attention.

SECTION 04 | CAREER SWOC ROADMAP

Strengths and Development Areas

Strengths

Technical Starting Point

Python and SQL provide a useful base for scripting, data handling, and software projects. Introductory C helps me understand core programming concepts and how programs are structured.

Communication and Leadership

These skills can help me explain technical work, coordinate tasks, listen to feedback, and contribute to group projects. They will matter when working with users as well as developers.

Curiosity and Initiative

I enjoy learning new things and have chosen a career direction early in my degree. This gives me time to explore the field, build projects gradually, and revise my plans as I gain experience.

Weaknesses and Development Areas

AI Foundations

Build understanding of data, probability, machine learning, and language-model concepts before relying on frameworks.

Applied Experience

Turn learning into visible work. Each project should describe its problem, design, implementation, tests, limitations, and lessons learned.

Software Delivery Skills

Strengthen Git and GitHub, APIs, databases, debugging, testing, and deployment alongside AI-specific skills.

Time in the Field

As a first-year student, I am at the beginning of my professional journey. Coursework, small projects, faculty guidance, and peer collaboration can help me gain experience steadily.

SECTION 04 | CAREER SWOC ROADMAP

Opportunities and Challenges

Opportunities

• Academic runway: build progressively deeper skills over four years instead of trying to learn every AI tool at once.

• Project-led learning: choose campus needs such as a canteen seat-availability estimator or college FAQ assistant; start small and test with users.

• Learning communities: use documentation, faculty guidance, coding clubs, open-source projects, workshops, and hackathons to learn with feedback.

• Interdisciplinary applications: explore AI in education, accessibility, health, finance, and campus services while learning each domain's needs.

• Professional exposure: seek collaboration, research opportunities, mentorship, and suitable internships as skills grow.

Challenges

• Fast-changing tools: learn underlying ideas and keep project dependencies and documentation current.

• Proving capability: build working software with readable code, tests, and thoughtful trade-offs.

• Trustworthy AI: evaluate answers, protect data, show sources where possible, and use human review for consequential actions.

• Managing breadth: balance CSE fundamentals, AI, projects, and coursework with a small number of weekly goals.

Response Plan

Choose a focused project, define success, test it with intended users, document what works and fails, and ask for feedback. This builds evidence while addressing reliability and time-management challenges.

SECTION 04 | CAREER SWOC ROADMAP

Action Plan: Closing the Gaps

AI Foundations | Years 1-2

Study programming, probability, data preparation, machine-learning basics, and LLM concepts. Explain key ideas in short notes.

Project Portfolio | Begin in Year 1; grow yearly

Build two small programming or data projects, then one AI application with a defined user problem and documented evaluation.

Software Engineering | Years 1-3

Use Git and GitHub, write readable code, test key behavior, work with APIs and databases, and deploy a small app.

Agent Engineering | Years 2-3

Add tool use and multi-step workflows only after a basic application works. Include permission boundaries, error handling, and human confirmation.

Industry Readiness | Years 2-4

Join technical events, collaborate with peers, seek feedback from faculty and practitioners, and apply for suitable internships.

Using Opportunities Well

- Build a small first version around a genuine user need.
- Include setup steps, screenshots or a short demo, tests, limitations, and next steps.
- Share learning responsibly; do not publish private data or credentials.
- Join a coding or AI community, contribute a small improvement, or take part in a team challenge when ready.

SECTION 04 | CAREER SWOC ROADMAP

Four-Year Milestones

Year 1 | 2026-27

Focus: programming and CSE fundamentals, including Python, SQL, C, Git, problem-solving, and core coursework.

Deliverables: two small programming or data projects, one clear README, and regular practice.

Readiness evidence: explain code, use version control, and show tested project work.

Year 2 | 2027-28

Focus: machine-learning basics, APIs, databases, backend development, and LLM application patterns.

Deliverables: build and deploy an AI application; evaluate examples and record failures.

Readiness evidence: connect a model to data and explain quality, cost, and privacy trade-offs.

Year 3 | 2028-29

Focus: agent workflows, tool use, retrieval, evaluation, cloud basics, security, and teamwork.

Deliverables: a bounded agent project with safeguards, logs, evaluation, and user feedback; seek practical exposure.

Readiness evidence: handle errors, permissions, and human review in a workflow.

Year 4 | 2029-30

Focus: advanced application design, reliability, deployment, specialization, and career preparation.

Deliverables: capstone-quality project, polished portfolio, internship experience if available, and interview preparation.

Readiness evidence: present a project end to end and discuss engineering decisions and limitations.

Milestone dates are planning estimates for a four-year degree beginning in 2026; adjust them to the official academic calendar and progress.

SECTION 04 | CAREER SWOC ROADMAP

Conclusion and Speaking Script

Key Takeaways

• My Python and SQL skills, introductory C knowledge, communication, leadership, and curiosity are a useful starting foundation.

• My next priority is to strengthen fundamentals and demonstrate them through projects, evaluation, and teamwork.

• Becoming an AI application and agent engineer is a step-by-step process: learn fundamentals, build useful projects, test carefully, and keep improving.

Three-Minute Speaking Script

Good morning everyone. I am a first-year Computer Science and Engineering student at St. Joseph's College of Engineering. My career goal is to become an AI application and agent engineer. This SWOC roadmap helps me understand my starting point and plan how I can grow during my degree.

I want to build AI applications and agents that solve practical problems. My current strengths include Python and SQL programming skills, basic knowledge of C, good communication and leadership skills, and an active interest in learning new things. Choosing this direction early gives me time to build a strong foundation.

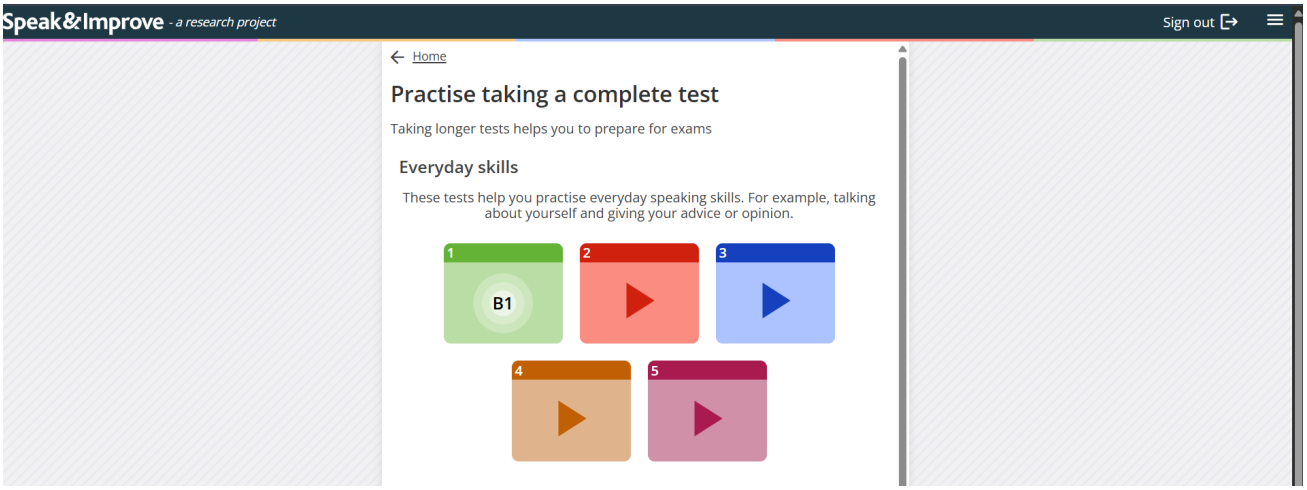
My development areas include learning AI and machine-learning fundamentals, building practical projects, and gaining experience with Git, APIs, testing, and deployment. I can develop these skills gradually through coursework and hands-on practice.

I have several years to explore this field through projects, college communities, workshops, open-source learning, and internships. At the same time, AI tools change quickly, competition is strong, and AI systems must be tested carefully for accuracy, privacy, and security. I will keep learning while maintaining a balance with core CSE subjects.

In my first year, I will strengthen programming fundamentals and complete small projects. In my second year, I plan to learn machine-learning and AI application basics and deploy an application. In my third year, I will explore agent workflows, evaluation, and cloud tools while seeking practical exposure. In my final year, I will focus on a capstone project, a strong portfolio, and career preparation. My main takeaway is that this career is built step by step: learn the fundamentals, build useful projects, test carefully, and keep improving. Thank you.

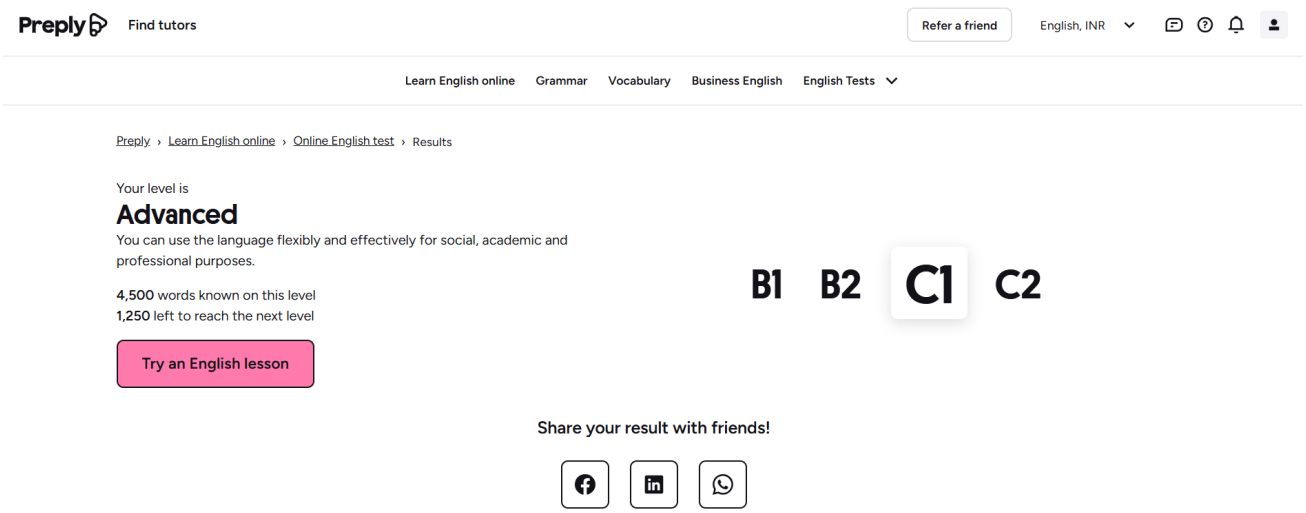
Write and Improve / Speak and Improve

Speak and Improve: score sheet (Task: Browsing on your phone)



Write and Improve / Speak and Improve

CEFR score card



My CEFR level is C1 (Advanced).

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must
5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully

10. (a) **Answer:** can

    (b) **Answer:** must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for every account.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with "https" and shows a padlock before entering any sensitive information.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or the National Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a

Bharatanatyam: The Dance That Carries Tamil Nadu's Soul

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

Question 28 (b) Formal letter of request to the Principal

Selected Question: 28.b

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal
St. Joseph's College of Engineering

OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,
Harshitha S
(Register No.: [Register No.])

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: [friend's email address]

Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Harshitha S

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.

2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in working condition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns and accidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors before disconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping the machine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor's permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and

night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Harshitha S
(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Harshitha S

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Harshitha S
(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹[amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval.

Thank you.

Yours obediently,
Harshitha S
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV

produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Harshitha S

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]

Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Harshitha S

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]

Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Harshitha S

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

SECTION 07 | PROJECT PROPOSAL: IDEATION

Smart Canteen Seat Availability System

C programming prototype | Harshitha S | Concept proposal

1. Overview

During busy breaks, students may spend valuable time walking around the canteen to find an empty seat. Seats may be available in less crowded areas that students do not know about, causing unnecessary crowding, wasted time, and frustration.

The Smart Canteen Seat Availability System is a proposed program that estimates and displays available seats across canteen zones. Its first version will be a simple C application that accepts or simulates occupancy counts, calculates free seats, and labels each zone Available, Filling up, or Busy. It demonstrates the core logic before sensor or mobile-app integration is attempted.

Problem Statement

Students lack a quick, shared view of seating availability by area. A clear estimate may help them choose where to go without identifying individuals or promising that a seat will remain free.

Project Purpose

Create an understandable C prototype that models zones and capacity, validates occupancy input, and reports updated availability. A small pilot could compare program output with manual counts at different times of day.

Project at a Glance

• Users: students seeking seating and canteen staff reviewing anonymous occupancy patterns.

• Language: C with standard library features and beginner-friendly functions.

• First version: console-based prototype with manually entered or sample counts.

• Outcome: a timely estimate by zone, not seat reservation or person tracking.

SECTION 07 | PROJECT PROPOSAL: IDEATION

2. Goals, Scope and Success Criteria

2.1 Goals

• Represent a canteen as named zones, each with a known seating capacity.

• Accept occupied-seat counts and calculate free seats without exceeding capacity.

• Display understandable labels: Available, Filling up, or Busy.

• Save and reload basic zone information using a plain-text or CSV data file.

• Test the program with sample cases and compare estimates with manual counts.

2.2 Non-Goals for the C Prototype

• No individual identification, facial recognition, or personal tracking.

• No live sensor hardware or guaranteed real-time occupancy in the first version.

• No seat reservation and no promise that an indicated seat will remain free.

• No direct SMS, WhatsApp, or push notifications; prototype alerts are shown in the terminal.

• No machine-learning prediction at the beginning; use transparent counts and rules.

2.3 Success Criteria

• Reject negative counts and occupied counts greater than zone capacity.

• For valid input, free seats equal capacity minus occupied seats.

• Users can distinguish all three status labels from the displayed result.

• Missing or malformed files produce a clear message and safe recovery.

• Record estimates and manual counts across several break periods for comparison.

2.4 Assumptions

For the first prototype, a trusted operator enters counts or loads sample data. The update interval, zone layout, and status thresholds must be reviewed with canteen staff before a live pilot.

SECTION 07 | PROJECT PROPOSAL: IDEATION

3. Users and Requirements

3.1 Intended Users

Students want to know which area is more likely to have seating before walking across the canteen. Canteen staff may enter or review counts and understand busy periods. The project team maintains the zone list, checks the program, and reviews pilot results.

3.2 Functional Requirements

• Add or load a zone with its name and seating capacity.

• Enter the occupied-seat count for a selected zone.

• Validate values and explain invalid input.

• Calculate free seats and a status label using agreed thresholds.

• Show all zones in a summary and allow a single-zone detail view.

• Save the latest counts and load them again later.

• Record the time of the latest update when available.

3.3 Quality Requirements

• Usability: clear menu choices, short prompts, and readable output.

• Reliability: invalid data must not corrupt counts or crash the program.

• Privacy: store zone totals only; do not collect names, IDs, or images.

• Transparency: describe results as estimates and show when each count was updated.

• Maintainability: keep calculations separate from input, display, and file operations.

3.4 Example User Stories

• Student: see which zones are Available or Busy to choose where to go.

• Staff member: update a zone count quickly so the estimate stays useful.

• Project team: compare estimates with manual observations to assess accuracy.

SECTION 07 | PROJECT PROPOSAL: IDEATION

4. C Technology and Data Model

A standard-library prototype with transparent calculations

4.1 Technology Choices

• Language: C (select C11 or another compiler-supported standard with the instructor).

• Libraries: standard headers such as stdio.h, stdlib.h, string.h, and time.h.

• Interface: console menu and text-based zone summary.

• Data: CSV or delimited text containing zone name, capacity, occupied count, and update time.

• Build and compiler: use the college lab compiler; document the chosen compile command.

• Testing: manual test cases and small checks for calculation functions.

4.2 Core Data Record

Each zone can be represented with a small structure. Limit text length, check every input, and reject counts outside the zone capacity.

typedef struct { char name[32]; int capacity; int occupied; } Zone;

4.3 Main Program Modules

• Input and menu: display actions and read validated choices.

• Zone management: load or update zone records.

• Availability logic: compute free seats and determine status from thresholds.

• Storage: read and write a plain-text file with clear error handling.

• Report display: show a summary and the timestamp or age of each count.

4.4 C Design Principles

Use small functions with clear inputs and outputs, constants for shared limits, and descriptive names. Keep first-release program state in bounded arrays. Check string length and numeric conversions before changing records.

SECTION 07 | PROJECT PROPOSAL: IDEATION

5. Availability Rules and Core Algorithm

Keep the first estimate explainable and easy to test

5.1 Status Rules

Initial thresholds are working assumptions and should be reviewed with users. For example: Available when at least 40% of seats are free; Filling up when more than 0% and less than 40% are free; Busy when no seats are free. Always show the numeric estimate with the status.

5.2 Calculation Sequence

1. Read zone capacity and occupied count from operator input or sample data.
2. Validate that capacity is positive and occupied is between zero and capacity.
3. Calculate free_seats = capacity - occupied.
4. Calculate free-seat percentage and apply the agreed thresholds.
5. Display the zone, free seats, approximate percentage, status, and update time.
6. Save valid updates for the next report.

5.3 Pseudocode

for each zone: validate capacity and occupied count; calculate free seats; classify the zone; print its summary and update time.

5.4 Stale Information

A count becomes less reliable as time passes or people move. Every display should show its update time. A later version may mark old readings “Stale” after a chosen time limit rather than presenting them as current.

5.5 Example

A zone with 40 seats and 22 occupied seats has 18 free seats (45%). Under the example thresholds, its status is Available. This is a sample calculation, not a live measurement.

SECTION 07 | PROJECT PROPOSAL: IDEATION

6. Data Handling and Error Management

Protect count integrity and make failures understandable

6.1 Data Fields

• zone_name: readable area label; required and limited in length.

• capacity: known seats in the zone; positive integer within the prototype limit.

• occupied: current count; integer from zero through capacity.

• updated_at: time when entered; readable timestamp or “unknown.”

6.2 File Operations

• Check that the data file opened successfully before reading or writing.

• Read one record at a time and validate separators, number conversion, and field lengths.

• When practical, write updates to a temporary file, then replace the old file only after a successful write.

• If the file is missing, explain how sample data can be loaded or a new file created.

• Never silently convert malformed values into apparently valid seat counts.

6.3 Input Validation

Read input into a bounded character buffer, convert numeric fields with a checked routine, and detect trailing invalid characters. If input is invalid, explain why and ask again without changing the saved zone.

6.4 Privacy and Access

The prototype uses anonymous zone-level counts only. It does not capture faces, names, student IDs, or device identifiers. If a future pilot adds hardware or shared access, first agree on permissions, retention, accuracy, and who can update counts.

SECTION 07 | PROJECT PROPOSAL: IDEATION

7. User Interaction and Notification Concept

A console-first version that could later support a dashboard

7.1 Console Workflow

• Display a welcome message explaining that availability is an estimate.

• Offer menu actions: view all zones, update a count, view one zone, reload data, or exit.

• After an update, show the zone name, free-seat estimate, status, and time.

• If a value is invalid, explain the acceptable range and keep the previous valid count.

7.2 Illustrative Output

Main Hall: 18 / 40 free (45%) | Available | updated 12:20 p.m.

Window Side: 5 / 24 free (21%) | Filling up | updated 12:19 p.m.

Outdoor Area: 0 / 16 free (0%) | Busy | updated 12:18 p.m.

Sample values only; not live canteen measurements.

7.3 Notification Concept

For the C prototype, a “notification” is an on-screen terminal message such as “Main Hall is now Available.” A later web or mobile client could show zone status or offer an optional alert. Any future alert must explain that counts may change before arrival.

7.4 Accessibility and Language

Do not rely on color alone. Use text labels and numbers, concise prompts, and readable terminal output. A future visual dashboard should be checked for readable type, contrast, keyboard access, and language clarity with students.

SECTION 07 | PROJECT PROPOSAL: IDEATION

8. Testing, Evaluation and Development Plan

Build the logic first, then evaluate with real observations

8.1 Test Cases

• Empty zone: occupied is 0; free equals full capacity; status is Available.

• Full zone: occupied equals capacity; free is 0; status is Busy.

• Partial count: valid input yields correct free-seat number and threshold label.

• Negative count: reject it and leave the saved value unchanged.

• Over-capacity count: reject it with a clear explanation.

• Missing file: show recovery options without crashing.

• Malformed record: reject the invalid entry and report the issue.

8.2 Pilot Evaluation

Select one canteen zone and compare the system estimate with manual counts at several peak and non-peak times. Record the estimate, observed count, time gap, and input mistakes. Review average absolute count difference and how often the displayed status matched the observed crowding level.

8.3 Development Phases

• Define: confirm zones, capacity, thresholds, and user needs; prepare sample data.

• Build: write C structures, input validation, calculation functions, and console report.

• Store: add checked file loading and saving; test malformed data.

• Evaluate: run sample cases and compare with manual observations.

• Improve: refine prompts, thresholds, and display based on user feedback.

SECTION 07 | PROJECT PROPOSAL: IDEATION

9. Risks, Limitations and Future Scope

Keep clear what the prototype can and cannot promise

9.1 Risks and Mitigations

• Counts become outdated: show update time; mark old readings stale in a later release.

• Manual entry introduces mistakes: validate ranges, provide clear prompts, and compare with observations.

• Status is mistaken for a guarantee: label it an estimate; do not reserve seats or promise availability.

• Scope grows too large: complete the console C prototype before adding sensors or mobile integration.

• Privacy concerns: store anonymous zone totals and do not identify or track individuals.

• Data is missing or corrupted: report errors and provide a clearly labeled recovery option.

9.2 Limitations

The first version depends on manual or sample inputs and cannot guarantee live counts. It does not model people moving between areas, seat reservations, or queue length. Its usefulness depends on count accuracy and freshness.

9.3 Future Scope

• Add a privacy-reviewed sensor or staff update workflow after evaluating the C program.

• Create a web or mobile dashboard that reads zone-level status from a secure service.

• Use historical anonymous counts to study peak periods and improve refresh timing.

• Offer optional alerts in a future client, with freshness labels and opt-in controls.

Conclusion

This project turns a familiar campus inconvenience into a manageable programming idea. A C prototype can establish the data model, validation, calculation, and reporting rules before live sensing is considered. Starting small makes it testable, transparent, and suitable for gradual improvement.

SECTION 08 | PROFESSIONAL PROFILES

LinkedIn Profile

Harshitha S | CSE Student | Aspiring AI Application and Agent Engineer

Suggested Headline

CSE Student at St. Joseph's College of Engineering | Aspiring AI Application and Agent Engineer | Python, SQL, C | Building Skills in Programming

About

I am a Computer Science and Engineering student at St. Joseph's College of Engineering, with a strong interest in artificial intelligence and software development. My career goal is to become an AI Application and Agent Engineer, building intelligent systems that address practical, real-world needs.

I am developing a solid foundation in programming, software engineering, and machine learning, while exploring how AI models, tools, and data can be combined to create useful applications and agents. I have skills in Python and SQL and basic knowledge of C. I value continuous learning, thoughtful problem-solving, and collaboration.

I look forward to applying my skills through projects and learning opportunities in AI engineering, and to connecting with professionals advancing practical, human-centered AI.

Education and Current Experience

• Education: St. Joseph's College of Engineering | Computer Science and Engineering | 2026-2030.

• Experience: Student, St. Joseph's College of Engineering | August 2026-present.

• Location: Chennai, Tamil Nadu, India.

• Skills shown: Canva, Microsoft PowerPoint, Engineering; developing Python, SQL, C, and DSA skills.

SECTION 08 | PROFESSIONAL PROFILES

LinkedIn: Skills and Profile Development

A clear, evidence-based profile that can grow with coursework and projects

Skills to Feature as They Are Practiced

Current foundation: Python, SQL, introductory C, communication, leadership, Canva, Microsoft PowerPoint.

Developing next: data structures and algorithms, Git and GitHub, C programming, APIs, databases, testing, AI/ML fundamentals, and responsible AI.

Evidence to add over time: course projects, code samples, short demos, documentation, team contributions, and certificates only after completion.

Profile Sections to Maintain

• Headline: state current study area and target domain clearly.

• About: describe interests, current skills, and the problems you want to work on.

• Education: keep college, degree, department, and expected graduation period accurate.

• Projects: add completed or active work; explain your role and what you learned.

• Skills: list skills you can discuss honestly and update them as your ability grows.

Professional Direction

My professional profile presents me as a first-year CSE student focused on programming and a future in AI application and agent engineering. My next step is to make this direction visible through consistent learning, practical projects, and clear documentation.

Contact

Email: harshithasubii@gmail.com | LinkedIn: linkedin.com/in/harshitha-s-a6aa91439 | GitHub: github.com/harshithasubii0709

SECTION 08 | PROFESSIONAL PROFILES

GitHub Profile and Next Steps

A learning profile that can show programming progress over time

GitHub Profile Snapshot

• Username: harshithasubii0709; profile name: Harshitha S.

• Bio shown: student at St. Joseph's College of Engineering; CSE; aspiring software engineer building programming skills.

• The supplied screenshot shows no public repositories yet and one contribution in the displayed last-year view.

Suggested GitHub Profile README

Hi, I'm Harshitha S, a first-year Computer Science and Engineering student at St. Joseph's College of Engineering.

Interested in AI applications, intelligent agents, and software that solves practical problems.

Currently learning: Python, SQL, C programming, data structures, and AI fundamentals.

Contact: harshithasubii@gmail.com | LinkedIn: linkedin.com/in/harshitha-s-a6aa91439

Repository Plan

• Publish a small C programming project after it is complete and reviewed.

• Add a README with purpose, requirements, build/run steps, examples, and known limitations.

• Use meaningful commit messages and keep secrets, personal data, and generated files out of public repositories.

• Pin only repositories that are ready for others to read; add screenshots or a short demo when useful.

Near-Term Career Roadmap

• Continue practicing programming fundamentals and SQL.

• Complete and document the C canteen availability prototype.

• Build a small Python project, then progress toward a carefully evaluated AI application.

• Update LinkedIn and GitHub with honest, verifiable evidence as skills grow.